Bayesian portfolio optimization: a comparison with the Plug-in approach

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1 Introduction

This report represents the implementation and comparison of two portfolio optimization approaches: the classical plug-in Markowitz strategy and a Bayesian framework based on a Normal-Inverse-Wishart prior.
Following the framework proposed by Lai, Xing, and Chen (2011) in *Mean–variance portfolio optimization when means and covariances are unknown* [1], this project takes their contribution as a starting point to investigate portfolio optimization under parameter uncertainty.
Rather than providing a full replication of the stochastic optimization approach proposed in the paper, the analysis focuses on a comparison between the classical plug-in estimator and a Bayesian framework, which introduces shrinkage through the use of conjugate priors.
The empirical analysis follows a standard workflow. It begins with the selection of a specific set of financial assets and their statistical analysis. The two portfolio strategies are then constructed, with particular attention to the estimation of expected returns and covariance matrices.
Finally, the approaches are evaluated and compared within an out-of-sample framework, in order to assess the potential benefits of Bayesian methods over classical plug-in estimators under different economic conditions.

2 Data analysis and preprocessing

2.1 Asset selection

The selection of assets plays a crucial role, as it allows the evaluation of portfolio performance under different market conditions. Rather than focusing on a single economic sector, a diversified set of financial assets is considered over a long time horizon (2000–2025). This extended sample includes periods characterized by varying levels of market uncertainty, such as the Global Financial Crisis (2008–2009), the COVID-19 pandemic (2020), and the subsequent inflationary phase.
The key idea behind this selection is not only to compare the statistical and technical differences between the two approaches, but also to identify market conditions under which one method may be preferred over the other.

2.2 Dataset description

The dataset consists of 20 assets covering multiple sectors. These include technology with well-known companies such as Apple, Microsoft, or financial with important firms like JPMorgan and Goldman Sachs, up to consumer goods as Coca-Cola, Walmart.
This diversification is relevant in terms of economic exposure, as different industries are affected by distinct economic drivers. For example, technology stocks tend to be more growth-oriented and sensitive to innovation cycles, while financial and energy sectors are more exposed to interest rates and price fluctuation.
No missing values were detected over the selected sample period, ensuring consistency in the subsequent analysis.

2.3 Log-return construction

After a first check of the correct sorting of the observations by date to ensure consistency in the analysis, the returns were constructed.

Portfolio optimization is performed using log-returns, computed as:

$$r_t = \log\left(\frac{P_t}{P_{t-1}}\right) \tag{1}$$

where P_t denotes the price of the asset at time t.
This standard transformation in finance field ensures time-additivity and statistical properties.

3 Exploratory Data Analysis

An initial exploration was conducted to prepare the dataset for the application of the different techniques and to assess whether the assumption of normality distribution of a classical portfolio implementation was satisfied.

3.1 Summary statistics and distributional properties

The main descriptive statistics reported for each asset are:

- The mean and standard deviation, providing information on the average return and volatility of each asset.
- Skewness, capturing the asymmetry of the return distribution.
- Excess kurtosis, measuring the heaviness of the tails relative to the normal distribution.

The results show that several assets exhibit non-zero skewness and significant excess kurtosis. This indicates that the return distributions deviate from Gaussianity, suggesting that the normality assumption underlying the classical mean-variance framework should be interpreted as an approximation.
In practice, this may contribute to instability in the estimation of mean and covariance parameters, especially in the presence of extreme observations.

3.2 Normality tests

To formally assess the normality assumption, the Shapiro–Wilk test was performed for each asset return series.
Let $\{r_{i,t}\}$ denote the return series of asset i. The Shapiro–Wilk test evaluates the null hypothesis:

$$H_0: \quad r_{i,t} \sim \mathcal{N}(\mu_i, \sigma_i^2) \tag{2}$$

against the alternative hypothesis of non-normality.
For each asset, the corresponding p-value was computed and the null hypothesis of normality rejected when

$$\text{p-value} < 0.05 \tag{3}$$

The empirical results indicate that, for the majority of assets, the null hypothesis is rejected with high significance levels, confirming that asset returns deviate from normality.
However, in the presence of non-normal returns, these estimators may be unstable and sensitive to extreme observations.
This provides a strong motivation for considering Bayesian approaches, which can incorporate prior information and reduce estimation risk.

3.3 Correlation and covariance structure

To understand the dependence between all the assets, the correlation and covariance matrices are computed.

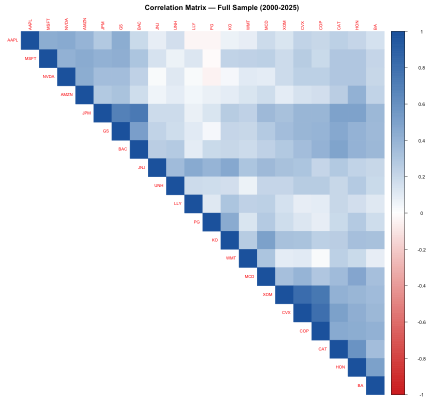

Figure 1: Correlation Matrix — Full Sample (2000–2025)

As shown in the figure, the majority of correlations are positive, with most pairs showing values between 0.2 and 0.7, which could indicate a common market component affecting all the assets. From a portfolio construction perspective, this structure has important implications. In particular, high correlations increase the sensitivity of the covariance matrix estimation, making the classical plug-in approach more unstable and subject to small changes. Secondly, particularly high correlations are observed within sectors:

- Financial stocks (e.g., JPM, GS, BAC) exhibit very high correlations, often above 0.6–0.7.
- Energy stocks (e.g., XOM, CVX, COP) are also highly correlated.
- Technology stocks (e.g., AAPL, MSFT, NVDA) show moderately high correlations, typically between 0.4 and 0.6.
- Healthcare stocks (e.g., JNJ, UNH, LLY) and industrial stocks (e.g., CAT, HON, BA) show moderate within-sector correlations, generally between 0.3 and 0.5.

The covariance matrix also plays a central role in portfolio optimization, as it directly determines portfolio risk:

$$\sigma_p^2 = w'\Sigma w$$

where Σ is the covariance matrix and w is the vector of portfolio weights.
When correlations are high, small estimation errors in Σ can lead to large changes in optimal weights. This further motivates the use of Bayesian approaches, which introduce shrinkage and improve the robustness of the covariance estimation.

4 Methodology

4.1 Mean-variance framework

Based on the mean–variance framework discussed in [1], a standard portfolio allocation problem, in which an investor distributes wealth across m risky assets, was considered.

Denoting $R_t \in \mathbb{R}^m$ the vector of asset returns at time t, with expected value $\mu = \mathbb{E}[R_t]$ and covariance matrix $\Sigma = \mathrm{Cov}(R_t)$, the investor chooses portfolio weights $w \in \mathbb{R}^m$ to balance expected return and risk by solving the mean-variance optimization problem:

$$\max_w \quad w'\mu - \lambda w'\Sigma w$$

where $\lambda > 0$ is the risk aversion parameter. In the empirical analysis it was set $\lambda = 5$, in line with one of the intermediate values considered in the simulation study of [1].
This formulation is known as the *Markowitz portfolio optimization* problem, where the first term represents the expected portfolio return, while the second term captures portfolio variance, the measure of risk. In practice, μ and Σ are unknown and must be estimated from the data, introducing the risk of estimation.
The key idea of this project is to compare both plug-in and Bayesian estimators to assess if there are particular economic and financial situations that require the specific use of one method instead of the other one.
It is worth noting that the Bayesian approach adopted here follows the conjugate estimation framework of Section 2.2 of [1], where posterior estimates of μ and Σ are used as inputs to the mean-variance optimization problem. This differs from the stochastic optimization approach proposed in Section 3 of the same paper (the NPEB method), which incorporates parameter uncertainty directly into the objective function via an additional parameter η.

4.2 Plug-in estimator

The first approach to Mean-Variance optimization is the *Plug-in* method. Since μ and Σ in Markowitz's efficient frontier are actually unknown, a natural idea is to replace them by their sample counterparts:

$$\hat{\mu} = \frac{1}{n}\sum_{t=1}^{n} r_t \tag{4}$$

$$\hat{\Sigma} = \frac{1}{n-1}\sum_{t=1}^{n}(r_t - \hat{\mu})(r_t - \hat{\mu})'. \tag{5}$$

The plug-in approach is highly sensitive to estimation error, especially in the mean vector $\hat{\mu}$. Small errors in expected returns can lead to large variations in optimal portfolio weights, resulting in unstable and poorly diversified portfolios.
To ensure numerical stability, a small regularization term is added to covariance matrix in the optimization step, preventing near-singularity issues.

4.3 Bayesian Normal-Inverse-Wishart model

To account for estimation risk, a Bayesian framework is adopted, and asset returns are assumed to be independently and identically distributed as:

$$r_t \sim \mathcal{N}(\mu, \Sigma)$$

A conjugate Normal-Inverse-Wishart prior is used:

$$\mu \mid \Sigma \sim \mathcal{N}(\mu_0, \Sigma/\kappa_0), \qquad \Sigma \sim \mathcal{IW}(\Psi, \nu_0)$$

where μ_0 is the prior mean, κ_0 controls the strength of belief in the prior mean, and Ψ, ν_0 govern the prior distribution of the covariance matrix.
The prior mean μ_0 corresponds to the parameter ν in the notation of [1].This change in notation is adopted to avoid confusion between the prior mean and the degrees of freedom parameter of the inverse-Wishart distribution.
Throughout the report, m denotes the number of assets, consistent with the notation of [1]. In the accompanying code, the same quantity is referred to as `p`, following standard R convention for the dimension of a multivariate distribution.
The degrees of freedom are set to $\nu_0 = m + 20$, where the minimum admissible value is $\nu_0 = m+1$. The additional 20 degrees of freedom introduce a weakly informative prior on the covariance structure, ensuring a finite prior mean while avoiding an overly diffuse specification. Note that ν_0 here corresponds to n_0 in the notation of [1], and to `nu0` in the accompanying code. Similarly, κ_0 corresponds to κ in [1]; the subscript is added here to distinguish it from the posterior quantity $\kappa_n = \kappa_0 + n$.
In the empirical implementation, the prior scale matrix Ψ is specified using an empirical Bayes approach. In particular, it is calibrated using the average variance observed within each estimation window, allowing the prior to adapt to different volatility regimes over time rather than being fixed ex-ante.
The prior mean is set to $\mu_0 = 0$, reflecting a neutral stance consistent with weak-form market efficiency: in the absence of prior information, expected returns are assumed to be zero, and the data pull the posterior mean away from this default.
As in the plug-in case, a small regularization term ϵI with $\epsilon = 10^{-6}$ is added to Σ_{pred} to ensure numerical stability of the quadratic programming solver.

4.4 Posterior predictive moments

Given a sample of returns $\{r_1, \ldots, r_n\}$, let $\bar{r}$ denote the sample mean and S the sample covariance matrix.
Under the Normal-Inverse-Wishart prior, the posterior parameters are:

$$\kappa_n = \kappa_0 + n, \quad \nu_n = \nu_0 + n,$$

$$\mu_n = \frac{\kappa_0 \mu_0 + n\bar{r}}{\kappa_n},$$

and

$$S_n = \Psi + (n-1)S + \frac{\kappa_0 n}{\kappa_n}(\bar{r} - \mu_0)(\bar{r} - \mu_0)'.$$

where S denotes the sample covariance matrix with denominator $n-1$, so that $(n-1)S = \sum_{t=1}^{n}(r_t - \bar{r})(r_t - \bar{r})'$ is the raw sum of squared deviations.

Posterior predictive covariance In this implementation, portfolio optimization is based on the predictive distribution of future returns.
Under the Normal-Inverse-Wishart model, the predictive distribution of r_{n+1} is a multivariate Student-t, whose covariance matrix is given by:

$$\Sigma_{\text{pred}} = \frac{\kappa_n + 1}{\kappa_n(\nu_n - m - 1)} S_n.$$

This choice is consistent with the framework of [1], which emphasizes that portfolio optimization should be based on the distribution of future returns rather than on point estimates of unknown

parameters. In this way, both covariance uncertainty and mean estimation uncertainty are incorporated into the risk measure.

4.5 Portfolio constraints

Both the plug-in and Bayesian portfolios are constructed under the same set of constraints. In particular, the optimization problem enforces:

- Full investment constraint: $\sum_{i=1}^{m} w_i = 1$
- No short-selling constraint: $w_i \geq 0 \quad \forall i$

4.6 Performance metrics

Portfolio performance is evaluated by using standard risk-adjusted metrics computed on monthly out-of-sample returns:

- Annualized return: $\mu_{ann} = 12 \cdot \bar{r}$
- Annualized volatility: $\sigma_{ann} = \sqrt{12} \cdot \sigma_r$
- Sharpe ratio: $SR = \mu_{ann}/\sigma_{ann}$
- Maximum drawdown: worst peak-to-trough decline of cumulative wealth

All metrics are computed on log-returns and reported for plug-in, Bayesian and equal-weight benchmarks.

4.7 Turnover and stability measures

Turnover: Portfolio turnover is defined as the average absolute change in portfolio weights between consecutive rebalancing periods:

$$\text{Turnover}_t = \sum_{i=1}^{m} |w_{i,t} - w_{i,t-1}|$$

and is averaged over time.

Equal-weight turnover:

Equal-weight turnover: For the equal-weight portfolio, turnover arises from price drift: after each period, asset weights deviate from the target $1/m$ because assets earn different returns. Turnover is therefore measured as the average absolute deviation between the drift-adjusted weights $\tilde{w}_{i,t}$ and the equal-weight target:

$$\text{Turnover}_t^{\text{EW}} = \sum_{i=1}^{m} \left| \tilde{w}_{i,t} - \frac{1}{m} \right|$$

where

$$\tilde{w}_{i,t} = \frac{w_{i,t-1}(1 + r_{i,t})}{\sum_{j=1}^{m} w_{j,t-1}(1 + r_{j,t})}$$

are the weights after price drift and before rebalancing.

Weight stability: Weight instability is measured as the standard deviation of asset weights over the out-of-sample period:

$$\text{Instability}_i = \text{sd}(w_{i,t})$$

Higher turnover implies higher transaction costs, while higher instability indicates less stable allocation strategies.

5 Backtesting framework

5.1 Rolling window design

In the empirical analysis, a rolling-window estimation framework was implemented to evaluate out-of-sample portfolio performance. At each time t, portfolio weights are estimated using the previous 60 months of data and then applied to the next period return. This procedure is repeated sequentially over the entire sample, allowing the portfolio to adapt itself to evolving market conditions.
The estimation window of $n = 60$ months is chosen following [1] (Section 5.2), who caution against settings where m is not small relative to n, yielding a ratio $n/m = 3.0$ while preserving a long enough out-of-sample period to include the Global Financial Crisis of 2008–2009.

5.2 Out-of-sample returns

The key idea of out-of-sample returns is to measure the future performance of portfolios constructed using only information available today.
At each monthly re-balancing date, the models parameters are re-estimated and the weights of the following portfolios are computed:

- **Plug-in portfolio**: classical mean-variance optimization using sample estimates
- **Bayesian portfolio**: shrinkage-based mean-variance optimization under a Normal-Inverse-Wishart prior
- **Equal-weight portfolio**: naive diversification strategy with constant weights $w_i = 1/m$

In this way, the portfolio weights are calculated at time t and then applied to future returns at $t + 1$, computing the out-of-sample returns.

5.3 Training sample weights

Portfolio weights are computed using the initial 60-month training sample to analyze the structural differences between the approaches.

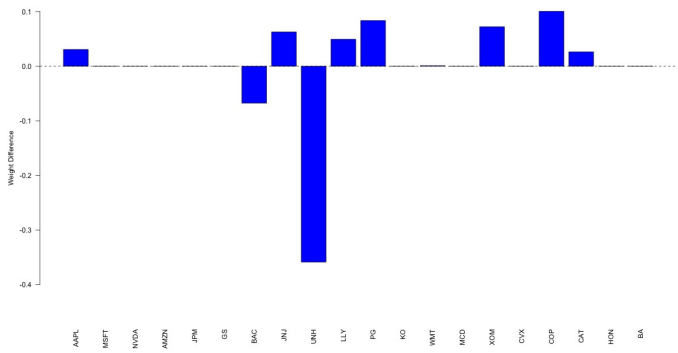

Figure 2: Difference in portfolio weights between Bayesian and plug-in estimator (training sample)

Positive values indicate that the Bayesian approach assigns a higher weight relative to the plug-in estimator, while negative values indicate a reduction in weight. A key feature of the results is the presence of a large negative difference for one asset (UNH), which receives a dominant allocation in the plug-in portfolio. The Bayesian approach substantially reduces this extreme position redistributing the weights.
Overall, the figure highlights how the Bayesian estimator mitigates the tendency of the plug-in approach to generate highly concentrated portfolios and generating more balanced portfolios, which is consistent with the shrinkage effect of the prior.
These results confirm the well-known instability of mean-variance optimization based on sample estimates, particularly for expected returns.

6 Empirical results

6.1 Cumulative wealth comparison

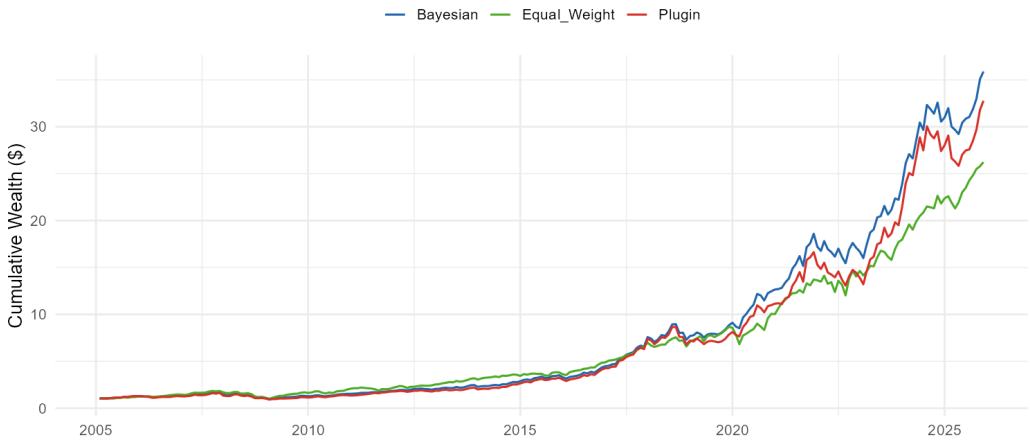

Figure 3: Out-of-sample cumulative wealth

The cumulative wealth trajectories provide a clear picture of how the strategies evolve over time. In the early part of the sample, the Equal-Weight portfolio performs slightly better, which is consistent with the idea that avoiding estimation can be beneficial in more stable conditions.

This advantage gradually disappears, and from the middle of the sample onwards both the Plug-in and Bayesian strategies begin to outperform. Between the two, the Bayesian portfolio tends to remain slightly above the Plug-in, with the gap becoming more visible over time. Rather than large differences at specific points, the effect appears to build gradually.
The shape of the trajectories also differs slightly: the Bayesian path appears somewhat smoother, while the Plug-in exhibits marginally more irregular movements. These differences remain, however, very limited, and both strategies follow similar dynamics.

6.2 Performance metrics

The following table summarises the main out-of-sample results.

Metric	Plug-in	Bayesian	Equal-Weight
Annualised Return (%)	16.68	17.12	15.62
Annualised Volatility (%)	16.95	15.17	15.55
Sharpe Ratio	0.984	1.129	1.005
Max Drawdown (%)	-43.91	-41.62	-46.08
Avg Turnover	0.220	0.165	0.047

Table 1: Out-of-sample performance comparison

The Bayesian strategy outperforms the Plug-in across all five metrics. It achieves a higher annualised return (17.12% versus 16.68%), lower volatility (15.17% versus 16.95%), a superior Sharpe ratio (1.129 versus 0.984), a smaller maximum drawdown (-41.62% versus -43.91%), and lower average turnover (0.165 versus 0.220). What stands out is not the size of any individual improvement, but the consistency across dimensions: the Bayesian prior regularises both the mean and covariance estimates, producing a more stable allocation that compounds its advantage over a 20-year out-of-sample period spanning multiple market regimes.

6.3 Drawdown analysis

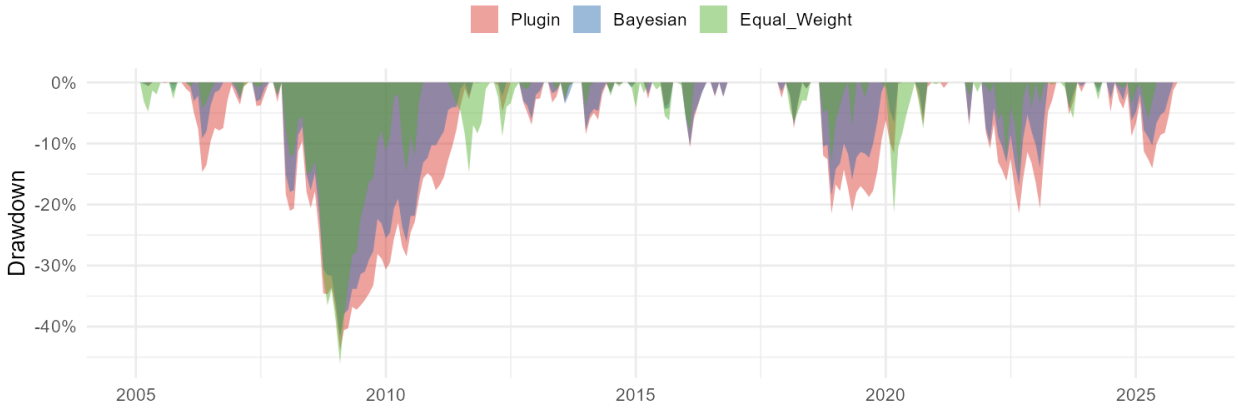

Figure 4: Drawdown comparison

The drawdown profiles highlight how the strategies behave during adverse periods. During the Global Financial Crisis, all portfolios experience substantial losses, but what really differs is not

when these losses occur, but how they accumulate over time.
The Plug-in strategy tends to remain in drawdown for longer periods, with wider and more persistent shaded areas, suggesting that losses are not only deeper but also harder to recover from. The Bayesian portfolio, on the other hand, does not necessarily recover faster, but seems to limit the overall build-up of losses, resulting in a trajectory that feels more contained and less exposed to prolonged stress.
The Equal-Weight benchmark behaves differently again: it occasionally reaches lower points, but these episodes tend to be shorter-lived. In this sense, the distinction across strategies is less about isolated extremes, and more about how losses persist and compound over time.

6.4 Downside risk analysis

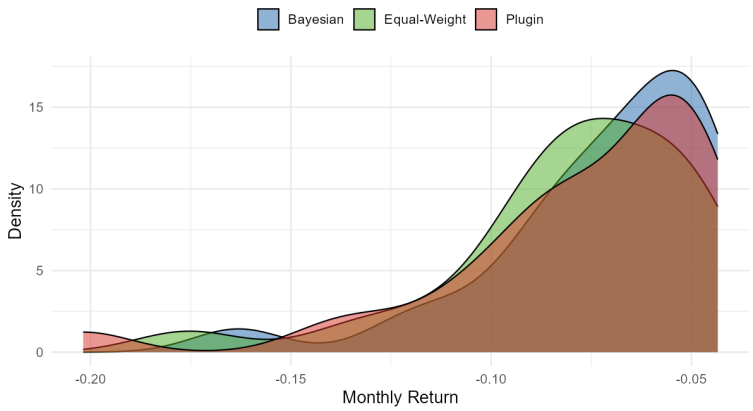

Figure 5: Left tail of portfolio returns (worst 10%)

Looking beyond averages and volatility, the left tail of the return distribution makes the differences more evident, as it isolates the behaviour of the strategies in the worst outcomes.
The Bayesian portfolio shows a higher concentration of observations in the upper part of the left tail, but this mass is shifted towards less negative values. Losses still occur, but they tend to be less extreme. The Plug-in strategy exhibits a similar pattern, although its distribution extends further to the left, indicating a higher exposure to more severe losses.
The Equal-Weight benchmark lies somewhere in between, with a more balanced shape but still a wider spread into negative territory.
What emerges quite clearly is that the main difference is not in how often losses occur, but in how severe they become. This is confirmed by the probability of large losses: returns below -5% are less frequent for the Bayesian strategy (7.2% of months versus 8.8% for the Plug-in), suggesting that its main contribution lies in containing the most extreme outcomes rather than avoiding losses altogether.

6.5 Turnover and weight instability

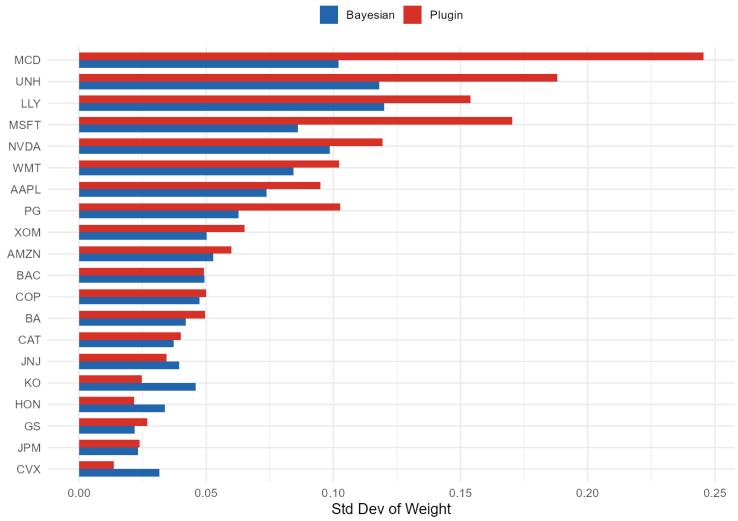

Figure 6: Weight instability: standard deviation of portfolio weights over time

The variability of portfolio weights highlights a different aspect of the comparison. The Plug-in strategy exhibits large fluctuations for several assets (most notably MCD, UNH and LLY) indicating that the optimizer reacts strongly to changes in the estimation window, producing unstable allocations over time. The Bayesian approach, in contrast, produces systematically lower weight variability across virtually all assets, a direct consequence of the NIW prior shrinking the posterior covariance toward a stable diagonal structure.
This reduction in weight instability is reflected in the lower average turnover reported in Table 1, confirming that the prior not only improves risk-adjusted returns but also stabilises the rebalancing behaviour of the portfolio.

7 Sensitivity analysis

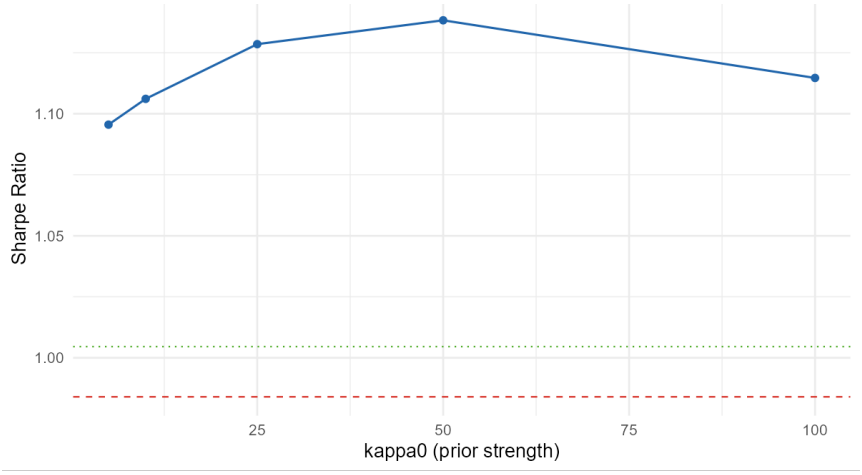

Figure 7: Sharpe ratio as a function of prior strength

Before settling on a specific value of κ_0, it is worth asking how much the results actually depend on it. The prior strength parameter controls how strongly the posterior mean is pulled toward zero: a small κ_0 leaves the model largely data-driven, while a large one increasingly dominates the likelihood. With a 60-month estimation window, $\kappa_0 = 25$ was chosen as a natural middle ground (equivalent to 25 pseudo-observations, enough to provide some regularisation without pretending to know more than the data). Varying κ_0 across $\{5, 10, 25, 50, 100\}$ shows that the Bayesian portfolio outperforms the Plug-in at every point in the grid, which is reassuring: the advantage does not appear and disappear depending on a single number, but holds up consistently as the prior is tightened or relaxed. The Sharpe ratio climbs from 1.096 at $\kappa_0 = 5$ to a peak of 1.138 at $\kappa_0 = 50$, then retreats slightly to 1.115 at $\kappa_0 = 100$, tracing out the familiar shape of the bias-variance trade-off.

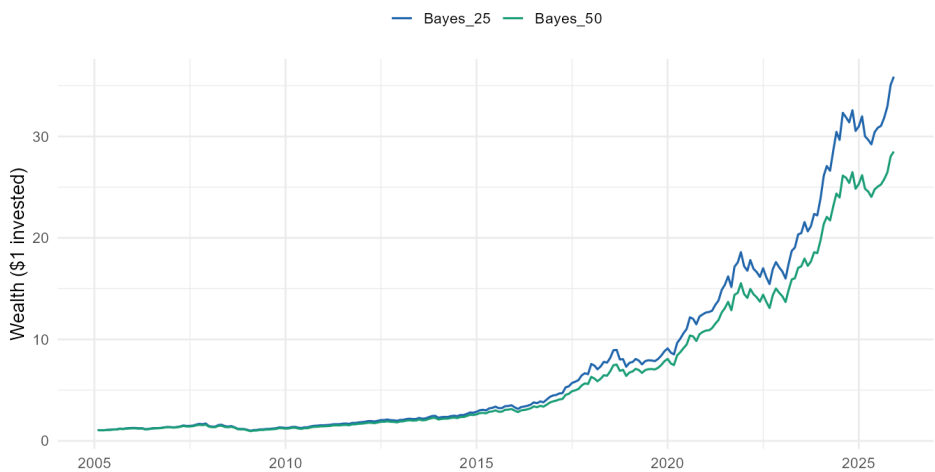

Figure 8: Cumulative wealth: $\kappa_0 = 25$ vs $\kappa_0 = 50$

The direct comparison between $\kappa_0 = 25$ and $\kappa_0 = 50$ tells a more nuanced story. For most of the sample the two specifications are nearly interchangeable, tracking each other closely through the pre-2020 period with only marginal differences in wealth accumulation. It is only after the pandemic recovery that a gap opens up, with $\kappa_0 = 25$ pulling noticeably ahead, a sign that the weaker prior, less anchored to its initial beliefs, was better positioned to adjust to the sharp shift in return dynamics that followed. Taken together, the evidence supports $\kappa_0 = 25$ on both theoretical and empirical grounds: it is a credible prior given the sample size, and it turns out to be the more adaptive choice when it matters most.

8 Regime analysis

Breaking the sample into different market phases provides additional perspective, although the differences remain relatively contained. The Bayesian approach tends to hold a small edge across most regimes, with the gap narrowing during stress periods such as the Global Financial Crisis and the inflationary phase of 2022, when all strategies deteriorate and move closer together.

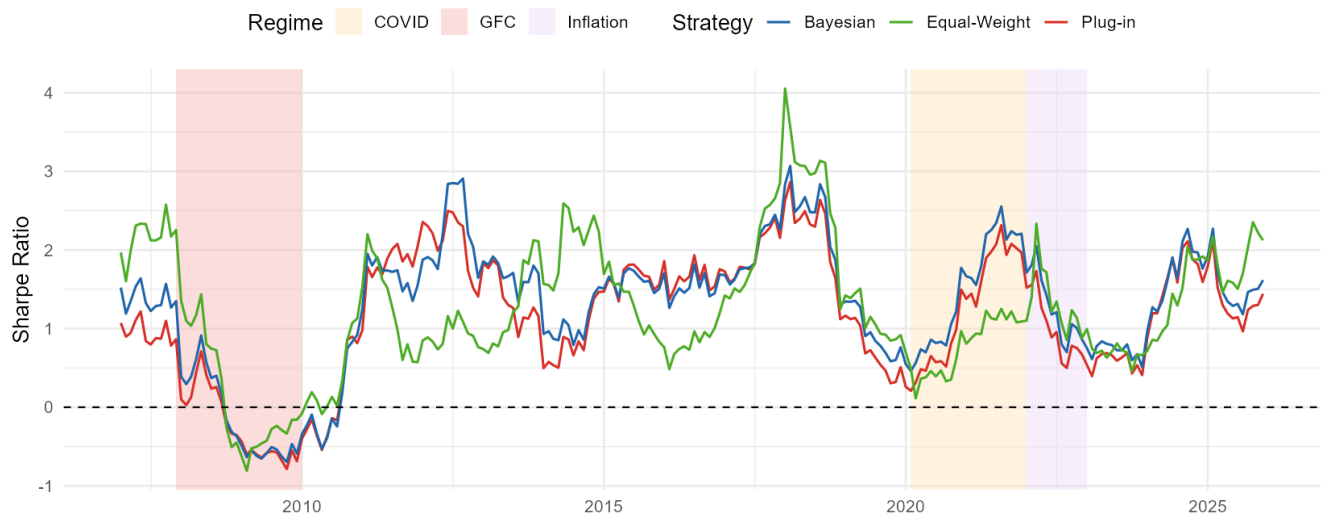

Figure 9: Rolling Sharpe ratio

The rolling Sharpe ratio confirms this picture. The Bayesian and Plug-in lines follow very similar paths throughout, with the Bayesian line sitting slightly above for most of the sample (most visibly during the recovery from the 2008 crisis and again after 2020). Around the COVID period, using a 24-month rolling window, both optimized strategies appear to hold up better than the Equal-Weight benchmark, which briefly turns negative. These results should be interpreted with caution: some regimes include relatively few observations, making annualised measures sensitive to noise. The regime analysis is therefore mainly descriptive, and it helps understanding how the strategies behave over time, but should not be read as definitive statistical evidence.

9 Discussion

At this point, the comparison between the Plug-in and Bayesian approaches is less about identifying a clear winner and more about understanding where the difference actually comes from. In many periods the strategies move closely together, reacting in similar ways to market conditions, and the gap never becomes dramatic.

The difference becomes clearer when considering how each method handles uncertainty. As [1] show, the plug-in approach ignores the variance of the estimated mean $\mathrm{Var}(w^\top \mu_n)$ in the decomposition of portfolio risk, effectively treating $\hat{\mu}$ as known, and this is precisely what causes it to overreact when estimates are noisy. The Bayesian approach introduces a form of moderation, smoothing these reactions without ignoring the data. This shows up most directly in the weight instability and turnover figures, where the Bayesian portfolio rebalances less aggressively across the board.

The cumulative effect of this moderation is visible in the performance metrics. The Bayesian strategy achieves a higher return, lower volatility, a better Sharpe ratio, a smaller maximum drawdown and lower turnover, showing a consistent improvement across all dimensions, even if no single gap is large on its own. The contribution lies as much in shaping the risk profile as in improving average performance.

These results should be interpreted with care, as the analysis relies on a specific dataset, ignores transaction costs, and assumes a long-only setting.

10 Conclusion

The evidence suggests that incorporating parameter uncertainty into portfolio optimisation can be beneficial, even when the improvement is not dramatic. The Bayesian approach does not revolutionise performance, but consistently edges ahead of the Plug-in across return, volatility, drawdown and turnover, a pattern that holds up across different market regimes and prior specifications.

The main mechanism is not a better forecast of future returns, but a more disciplined reaction to noisy estimates. By shrinking the posterior toward a stable prior, the model avoids the overreaction that characterises the Plug-in, producing allocations that change less abruptly from one period to the next. This effect is modest at any given point in time, but compounds over a long out-of-sample period.

The downside findings reinforce this interpretation. The Bayesian strategy does not reduce the frequency of losses, but tends to make them less severe, with a lower probability of extreme monthly outcomes. Its role is therefore more about containing tail risk than improving average performance.

In this case, the Bayesian framework proves to be a more robust alternative to the classical Plug-in approach, particularly when estimates are noisy and the estimation window is short relative to the number of assets. Whether this advantage generalises to other datasets, asset universes or time periods remains an open question.

Several extensions could further develop this framework. A natural next step would be to replace the Sharpe ratio with a Value-at-Risk or Expected Shortfall criterion, which would more directly target the tail risk reduction that the Bayesian approach appears to deliver in this case. Incorporating time-varying priors or dynamic estimation windows could also help the model adapt more quickly to structural breaks, such as those observed around the 2008 crisis and the 2020 pandemic shock.

References

[1] Lai, T. L., Xing, H., and Chen, Z. (2011). *Mean-variance portfolio optimization when means and covariances are unknown.* The Annals of Applied Statistics, 5(2A), 798–823.